Phase 1: Modelling, Equilibria and Linearization of the Acrobot

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Abstract. **We verify the given Acrobot model, find its equilibria for zero motor torque, linearize the dynamics at each equilibrium, and derive the transfer functions from the elbow torque τ to both joint angles. The linear models are checked against nonlinear simulation.**

I. Nonlinear Model

A. System and Parameters

Two-link pendulum (Fig. 1) with a passive shoulder and a motor at the elbow [1,2].

- Angles: q_1 = link 1 from the downward vertical; q_2 = link 2 relative to link 1; $\varphi = q_1 + q_2$.
- State: $x = [q_1,\ q_2,\ \dot{q}_1,\ \dot{q}_2]^T$.
- Input: elbow motor torque τ, $|\tau| \le 20\,\mathrm{N\,m}$, the only quantity we command ($\tau = 0$ in this phase; set by a controller in later phases).
- $m_1 = m_2 = 1$ kg, $l_1 = 1$ m, $l_2 = 2$ m, $l_{c1} = 0.5$ m, $l_{c2} = 1$ m, $I_1 = 0.083$, $I_2 = 0.33\ \mathrm{kg\,m^2}$, $g = 9.81\ \mathrm{m/s^2}$.

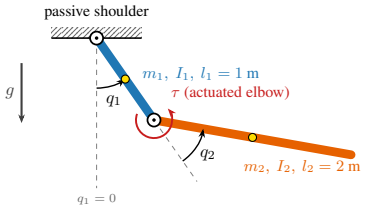


Fig. 1. Acrobot schematic (to scale); dots mark the centres of mass.

B. Given Equations of Motion

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + B\tau, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{1}$$

(B applies τ to the elbow equation only.) Below we derive each term with $c_2 = \cos q_2$, $s_2 = \sin q_2$.

C. Derivation

1) Potential energy and gravity torque:

$$\begin{aligned} U &= -(m_1 l_{c1} + m_2 l_1)\, g \cos q_1 - m_2 l_{c2}\, g \cos\varphi \\ &= -14.715 \cos q_1 - 9.81 \cos\varphi, \\ \tau_g &= -\frac{\partial U}{\partial q} = -9.81 \begin{bmatrix} 1.5 \sin q_1 + \sin\varphi \\ \sin\varphi \end{bmatrix}. \end{aligned} \tag{2}$$

2) Kinetic energy and inertia matrix: the velocity of centre 2 is the elbow velocity ($l_1\dot{q}_1$) plus the rotation about the elbow ($l_{c2}\dot{\varphi}$), at angle q_2 apart (law of cosines):

$$\begin{aligned} T &= \tfrac{1}{2}(m_1 l_{c1}^2 + I_1)\dot{q}_1^2 + \tfrac{1}{2} I_2 \dot{\varphi}^2 \\ &\quad + \tfrac{1}{2} m_2 \left(l_1^2 \dot{q}_1^2 + l_{c2}^2 \dot{\varphi}^2 + 2 l_1 l_{c2} c_2 \dot{q}_1 \dot{\varphi} \right) = \tfrac{1}{2} \dot{q}^T M \dot{q}, \end{aligned}$$

$$M(q) = \begin{bmatrix} 2.663 + 2c_2 & 1.33 + c_2 \\ 1.33 + c_2 & 1.33 \end{bmatrix}. \tag{3}$$

3) Coriolis term (Euler–Lagrange):

$$\begin{aligned} &\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \tau_g + B\tau, \qquad \frac{\partial T}{\partial \dot{q}} = M\dot{q} \\ &\Rightarrow\ M\ddot{q} + \underbrace{\left(\dot{M}\dot{q} - \frac{\partial T}{\partial q} \right)}_{= C\dot{q}} = \tau_g + B\tau. \end{aligned}$$

With $\dot{M} = \frac{\partial M}{\partial q_2}\dot{q}_2 = -s_2\dot{q}_2 \left[\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ this evaluates to

$$C(q,\dot{q}) = s_2 \begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}. \tag{4}$$

These are the same τ_g, M and C as in the given model.

D. State-Space Form

$$\begin{aligned} \det M &= (2.663 + 2c_2)(1.33) - (1.33 + c_2)^2 \\ &= 1.7729 - c_2^2 \ \ge\ 0.7729 > 0 \quad (c_2^2 \le 1), \\ \dot{x} = f(x,\tau) &= \begin{bmatrix} \dot{q} \\ M^{-1}\left(\tau_g + B\tau - C\dot{q}\right) \end{bmatrix}. \end{aligned} \tag{5}$$

II. Equilibria

A. Conditions

At rest $\dot{q} = \ddot{q} = 0$, and the input torque is given as $\tau^* = 0$. Then (1) gives $\tau_g(q^*) = 0$:

$$\begin{aligned} &\text{row 2:} \quad \sin(q_1^* + q_2^*) = 0, \\ &\text{row 1:} \quad 1.5 \sin q_1^* + \underbrace{\sin(q_1^* + q_2^*)}_{=0} = 0 \ \Rightarrow\ \sin q_1^* = 0. \end{aligned}$$

B. Solutions

$$q_1^* \in \{0, \pi\}, \quad q_1^* + q_2^* \in \{0, \pi\} \ \Rightarrow\ q_2^* \in \{0, \pi\} \pmod{2\pi}.$$

$\Rightarrow$ **four equilibria** (Table I), all with both links vertical.

TABLE I
Equilibria for $\tau^* = 0$

	Posture	(q_1^*, q_2^*)	U [J]	Description
E_1	down–down	$(0,0)$	-24.525	both links hanging
E_2	down–up	$(0,\pi)$	-4.905	link 1 down, link 2 up
E_3	up–up	$(\pi,0)$	$+24.525$	fully upright (target)
E_4	up–down	(π,π)	$+4.905$	link 1 up, link 2 down

TABLE II
TRANSFER FUNCTIONS $G_i(s) = N_i(s)/\Delta(s)$ AT THE FOUR EQUILIBRIA

	Denominator $\Delta(s)$ (monic)	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$	Poles	Zeros of G_1	Zeros of G_2
E_1	$s^4+42.2410s^2+186.7719$	$-3.0147(s^2+4.2103)$	$6.0332(s^2+5.2595)$	$\pm j2.2400,\ \pm j6.1011$	$\pm j2.0519$	$\pm j2.2934$
E_2	$s^4+8.4025s^2-186.7719$	$-0.4270(s^2-29.7273)$	$0.8578(s^2+7.3982)$	$\pm 3.1775,\ \pm j4.3010$	± 5.4523	$\pm j2.7200$
E_3	$s^4-42.2410s^2+186.7719$	$-3.0147(s^2-4.2103)$	$6.0332(s^2-5.2595)$	$\pm 2.2400,\ \pm 6.1011$	± 2.0519	± 2.2934
E_4	$s^4-8.4025s^2-186.7719$	$-0.4270(s^2+29.7273)$	$0.8578(s^2-7.3982)$	$\pm 4.3010,\ \pm j3.1775$	$\pm j5.4523$	± 2.7200

III. LINEARIZATION

A. Deviation Variables

$$\tilde{q} = \begin{bmatrix} q_1 - q_1^* \\ q_2 - q_2^* \end{bmatrix}, \quad \tilde{x} = \begin{bmatrix} \tilde{q} \\ \dot{\tilde{q}} \end{bmatrix} = x - x^*, \quad \tilde{\tau} = \tau - \tau^* = \tau.$$

First-order Taylor expansion of (5) about $(x^*, 0)$:

$$\dot{\tilde{x}} \approx \underbrace{f(x^*,0)}_{=0} + \left.\frac{\partial f}{\partial x}\right|_* \tilde{x} + \left.\frac{\partial f}{\partial \tau}\right|_* \tilde{\tau}.$$

B. Jacobians

Top half: $\frac{d}{dt}\tilde{q} = \dot{\tilde{q}} \Rightarrow$ rows $[0\ \ I_2]$.

Bottom half: $h = M^{-1}(\tau_g + B\tau - C\dot{q})$. At $(x^*, 0)$:

$$\partial(C\dot{q})/\partial\dot{q} = 0 \quad (C\dot{q} \text{ is quadratic in } \dot{q},\ \dot{q}^* = 0)$$
$$(\partial M^{-1}/\partial q)(\tau_g + B\tau) = 0 \quad (\tau_g(q^*) = 0,\ \tau^* = 0)$$

$$\Rightarrow \ddot{\tilde{q}} = P\tilde{q} + r\tilde{\tau}, \qquad P = M(q^*)^{-1}K(q^*), \quad r = M(q^*)^{-1}B,$$

$$K(q^*) = \left.\frac{\partial \tau_g}{\partial q}\right|_{q^*} = -9.81 \begin{bmatrix} 1.5\cos q_1^* + \cos\varphi^* & \cos\varphi^* \\ \cos\varphi^* & \cos\varphi^* \end{bmatrix}. \quad (6)$$

C. Linear State-Space Model

Stacking both halves, with the joint angles as outputs, $y = \tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$:

$$\boxed{\begin{aligned} &\dot{\tilde{x}} = A\tilde{x} + B_\ell\tilde{\tau}, \qquad y = C_y\tilde{x} + D\tilde{\tau}, \\ &A = \begin{bmatrix} 0 & I_2 \\ P & 0 \end{bmatrix},\ B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix},\ C_y = \begin{bmatrix} I_2 & 0 \end{bmatrix},\ D = 0 \end{aligned}} \quad (7)$$

D. Example: $E_1 = (0,0)$

$\cos q_1^* = \cos\varphi^* = c_2 = 1$:

$$K = \begin{bmatrix} -24.525 & -9.81 \\ -9.81 & -9.81 \end{bmatrix}, \quad M = \begin{bmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{bmatrix},$$
$$\det M = 0.7729, \qquad M^{-1} = \frac{1}{0.7729}\begin{bmatrix} 1.33 & -2.33 \\ -2.33 & 4.663 \end{bmatrix},$$
$$P_{E1} = M^{-1}K = \begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}, \quad r_{E1} = \begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}.$$

E. All Equilibria

M depends only on q_2^*; K only on the signs of $\cos q_1^*$, $\cos\varphi^*$. Upside down, gravity flips:

$$K_{E3} = -K_{E1},\ M_{E3} = M_{E1} \Rightarrow P_{E3} = -P_{E1}; \qquad P_{E4} = -P_{E2}.$$

Results in Table III; with (7) they give (A, B_ℓ, C_y, D) at each equilibrium.

TABLE III
LINEARIZATION RESULTS ($\det M = 0.7729$ AT ALL FOUR)

	$(\cos q_1^*, \cos\varphi^*)$	$M(q^*)$	$P = M^{-1}K$	$r = M^{-1}B$
E_1	$(1,1)$	$\begin{bmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{bmatrix}$	$\begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}$	$\begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}$
E_2	$(1,-1)$	$\begin{bmatrix} 0.663 & 0.33 \\ 0.33 & 1.33 \end{bmatrix}$	$\begin{bmatrix} -12.6292 & 12.6926 \\ 10.5095 & 4.2266 \end{bmatrix}$	$\begin{bmatrix} -0.4270 \\ 0.8578 \end{bmatrix}$
E_3	$(-1,-1)$	as E_1	$-P_{E1}$	r_{E1}
E_4	$(-1,1)$	as E_2	$-P_{E2}$	r_{E2}

IV. TRANSFER FUNCTIONS

A. Laplace Transform

Define $\tilde{Q}_i(s) = \mathscr{L}\{\tilde{q}_i(t)\}$, $\tilde{\mathscr{T}}(s) = \mathscr{L}\{\tilde{\tau}(t)\}$, $\tilde{Q} = [\tilde{Q}_1, \tilde{Q}_2]^T$. With zero initial conditions ($\ddot{\tilde{q}} \to s^2\tilde{Q}$):

$$s^2\tilde{Q} = P\tilde{Q} + r\tilde{\mathscr{T}} \ \Rightarrow\ (s^2I - P)\,\tilde{Q}(s) = r\,\tilde{\mathscr{T}}(s).$$

B. General Formula

$$(s^2I - P)^{-1} = \frac{1}{\Delta(s)}\begin{bmatrix} s^2 - P_{22} & P_{12} \\ P_{21} & s^2 - P_{11} \end{bmatrix},$$
$$\Delta(s) = \det(s^2I - P) = s^4 - \mathrm{tr}(P)\,s^2 + \det P.$$

Multiplying by r and dividing each row by $\tilde{\mathscr{T}}$ ($G_i = \tilde{Q}_i/\tilde{\mathscr{T}}$):

$$\boxed{\begin{aligned} G_1(s) &= \frac{r_1 s^2 + (P_{12}r_2 - P_{22}r_1)}{\Delta(s)}, \\ G_2(s) &= \frac{r_2 s^2 + (P_{21}r_1 - P_{11}r_2)}{\Delta(s)} \end{aligned}} \quad (8)$$

- *Poles:* roots of Δ (= eigenvalues of A); with $\lambda = s^2$, $\lambda^2 - \mathrm{tr}(P)\lambda + \det P = 0$, then $s = \pm\sqrt{\lambda}$.
- *Zeros:* roots of the numerators, $s^2 = -(P_{12}r_2 - P_{22}r_1)/r_1$ and $s^2 = -(P_{21}r_1 - P_{11}r_2)/r_2$.

C. Example: E_1

$$\mathrm{tr}\,P = -42.2410, \quad \det P = 186.7719$$
$$\Rightarrow \Delta = s^4 + 42.2410s^2 + 186.7719,$$
$$\lambda = \tfrac{-42.2410 \pm 32.206}{2} = -5.018,\ -37.224$$
$$\Rightarrow \text{poles} = \pm j2.2400,\ \pm j6.1011,$$
$$P_{12}r_2 - P_{22}r_1 = -12.6926, \quad P_{21}r_1 - P_{11}r_2 = 31.7316,$$

$$G_1 = \frac{-3.0147(s^2 + 4.2103)}{\Delta(s)}, \qquad G_2 = \frac{6.0332(s^2 + 5.2595)}{\Delta(s)}.$$

D. Results

Table II lists all four equilibria. With no friction only even powers of s occur, so poles and zeros come in pairs $\pm p$; since $P_{E3} = -P_{E1}$, each $\pm j\omega$ at E_1 becomes $\pm\omega$ at E_3 (likewise E_2/E_4).

V. Verification

- The model derivation and all values in Tables II and III were verified in MATLAB (agreement to four decimals).
- Nonlinear (5) vs. linear (7), simulated in Python from the same small deviation, $\tau = 0$ (Fig. 2): near E_1 from $\tilde{q}(0) = (0.1, -0.1)$ rad they agree within 1.0° (q_1) and 2.0° (q_2) over 8 s; near E_3 from $\tilde{q}_1(0) = 10^{-3}$ rad the error is 0.02% at $|\tilde{q}_1| = 1^\circ$.
- With $\tau = 0$, $T + U$ stays constant to 10^{-8} (no friction).
- **Animation** (Manim) of the model, the four equilibria, the nonlinear vs. linear responses near each equilibrium and the pole–zero maps: **watch the video**.

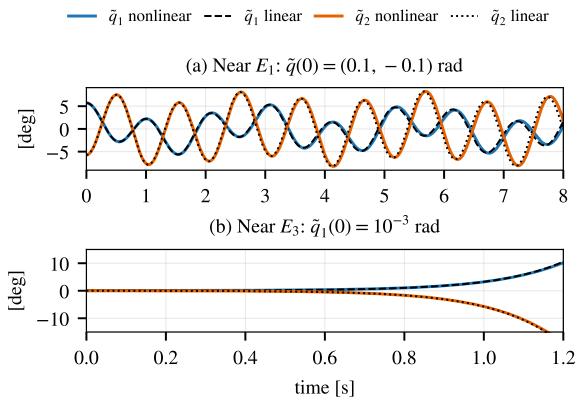


Fig. 2. Free responses of the nonlinear (colour) and linear (black) models near (a) E_1 and (b) E_3 (simulated in Python).

References

[1] M. W. Spong, "The swing up control problem for the Acrobot," *IEEE Control Syst. Mag.*, vol. 15, no. 1, pp. 49–55, 1995.

[2] R. Tedrake, *Underactuated Robotics*, MIT, 2024. [Online]. Available: https://underactuated.csail.mit.edu